ActInf Textbook Group ~ Cohort 8 ~ Session 9 (Chapter 5) ~ 4/17/2025

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All right, welcome back to cohort 8. We are definitely in the backstretch in the final chapter. So, we'll have two discussions on chapter 5. Then these last uh weeks, we tend to not record them. It's usually just like looking at the feedback form, giving giving people an opportunity to fill out the feedback form. I I'll update the numbers. Um, and then just looking at a lot of the thoughts people have had uh about future textbook groups and just as we get further and further from the initial book and sort of explore other reading groups. So, they're just fun and then that kind of closes this phase. then we'll probably resume with chapter six before uh in in some months but first let's go to oh and one other uh piece I just wanted to show initially Sanjie Nam Jooshi who's the author of the the soontobe published the fundamentals uh two textbooks with with one of them coming out um he has also made an open source repo that I'm just running it right now locally. And this is what it pops up from a Docker container. So that's kind of like uh veering from the textbook group, but all the different references and things that people bring up and and um just a and and it's also a cool place to explore like literature meta analysis and uh other kinds of literature based methods. Okay, chapter five though. Would anyone like to start with any any any recap of five or question or section from five uh I found it uh I'm sculowskas I found it very helpful um just in terms of making sense of active inference in terms of uh domain and maybe this is the domain where it uh was first uh recognized and formulated u and so uh uh which is neur neuroscience brain sciences and so for me personally personally like I like they make a note about the relating of the discrete models with the continuous models how that functions uh and so uh I think that there's something more to be discovered there and um I guess there's this theme of message passing so that's a code word I think for neurotransmitters saying that um uh you can model uh neurotransmission um with um Markoff processes and thus active inference becomes relevant and then there's also uh examples um with um the cortical columns in the cortex you know trying to make layers and how are those layers and so uh or uh I'm also interested like in the subcortical structures like the basil ganglia you have a excitatory uh I think direct pathway and inhibitory indirect pathway that are connecting with motor uh capabilities um sensory very helpful this idea that you need to with precision of the understanding I guess of the let's say sensors the proprio proprio centers I think um sensors like they need if if they think that they're not moving, they're not ever going to move. So you have to make them doubt whether they're static and then then you can let them think, oh, you're moving and then they start to move. Uh so that's a very interesting um dynamic. I think that that really helps to um I I think it it's just it helps to think about what's going on and then that makes active inference uh intuitively an attractive framework and then mathematically interesting. How how you know helpful. Thank you. Great comments. Anyone else have like a section or or a comment on five. I'll just say more like that. It really opens up a lot of things to think about. And I think also it makes clear that active inference has you know has something this very important idea that you know lost you for a few seconds and okay yeah two flavors of the same thing and so that there's three energy principle um can be understood in either you know can can get either one of those going. Um but um it also shows like that there's a lot there. It's just a it's just a tool for thinking about these things. But as far as actual implementation interpretation, there's a lot that it doesn't really um do for itself. You have to kind of like come up with a picture or a model or idea. And so one of the things that fascinated me was just appreciating oh that there's this whole language of neurotransmitters. Uh there's this whole um alphabet let's say of of possibilities. And uh I don't think active inference directly speaks to what they mean. It just kind of explains some of the framework for that language. So that's that's what I inferred. Thanks. Great comments. Yeah. Wow. There's a lot it doesn't do for itself. That's kind of a function of the the the generality. Um, and then like the table of neurotransmitters and these specific systems. This is where the sort of specific experimenting, hypothesis, testing, falsification, it's like we're not seeing any evidence from lesion studies or from this C elegance model and this mouse model. We're not seeing evidence for this cognitive function in connection to that one. So that's kind of a neuroarchchitecture agnostic way to frame it. Like we tried to apply this Baze model to these different and we found that there were these diff um these different structural models of parameters in play. And then it it could also connect with a more explicitly mechanistic research agenda like arising from SPM and the empirical work on fMRI and neuroiming where they take similar models to these and and look at how these parameters differ keeping the edge constant or look at how certain parameters differ um under different structural hypothe hypothesis like in attention or or um unattended context. That's kind of the SPM, fMRI, EEG classic studies which are kind of explored in chapter 9. So it's like chapter 5 is the first time that we're encountering the specific um cognitive in a continuum to neuromchanistic hypothesis falsifiable um drawn from larger possible hypotheses. Like if the even just keeping the same variables in play, there are other structural hypotheses that we could imagine. Um and and there might be a given area of phase space, could be small area, could be a huge area where like two structurally similar models are can't be one can't be said to be like better than the other like they're not identifiable from each other. And that that comes up with like structural equation modeling in just any kind of statistical method. [Music] Um, anyone else want to go to a a part where we could try to look at one of the papers that are cited for more information or look at one of the uh three subsystems more. Yeah, Sylvia, um, I do have a part that I've been thinking about, haven't really gotten my head around. is on page 95. So section 5.5 um at the bottom bit of that page it says the relationship between the two is simple. If we have very precise beliefs about how data are generated from hidden states then our beliefs about hidden states can be updated by observing data more than if we are not confident in those beliefs. I'm just wondering why isn't this the other way around? as in if you are less confident that you might have to update more that that same exact question. I'm glad you asked it. Yeah, it is a good question. So, um two two cases Um, one of them is the prior looks like a very broad Gaussian like it's um 10 plus or minus 5 and then a new observation comes in. So, so it's not a very precise belief. New observation comes in and kind of shifts the whole Gaussian slightly. That that's a low precision belief updating slightly. If there's a high precision belief, then it's like a a 10 plus or minus one. And if that same measurement of um 14 comes in, the the posterior could move a a lot to that. Now whether the wide uh high variance or the low variance prior moves not at all or all the way is like an attention precision question. So I think here um it just looking at all other things being equal basian and updating I think that um for the same attention the tight distribution would update possibly less because the the the observation is just so unlikely. So it's like the the posterior might not update that much but maybe because here they're focusing just on the precision but but it is separable from from see seemingly from like what you what the actual variance of the target distribution is whereas if you if you believe if you believe that it's an informative channel you update more and can I give an example and Do you see if it's Yeah. So, let's say I'm wondering where my brother is. So, one very vague idea, my brother went out. A less vague would be my brother went out to the store. And more precise would be my brother went out to 7-Eleven. So, you see, if he went out to 7-Eleven, it's easy to for that to turn out to be needing to be updated. You know, lots of things could say, well, he's he's on the other side of town, right? Like, so, but he's on the other side of town. He may still have gone to a store, right? So the more precise it is, the more easy it is vulnerable to, you know, being affected by more information. That's the way I'm I I think that's it's related. I I hope I'm not like o over splitting it, but it's like let's just go with a 7-Eleven case. So there's like the there's three people in the room you could ask um and they're going to tell you at those three levels of course graining of location like the 10's place, the one's place or the decimal place or all three. But then attention would be on a zero to 100 or 0 to one. How much are you going to update to that belief including not not mattering at all? Like if you already believed that he went out, then zero or 100 attention to the person who's going to tell you just that isn't going to update your belief at all, whether you're paying attention to that first person or not, because it's like the ball is already at the bottom of the bowl. It's it's already at the maximum likelihood solution. Whereas, um, if you had it 100% belief, you're going to update to just whatever the person tells you. So that's the P. So it's like the the the or just like the the propriception coming from a hand like that little corpsical or or unit could be firing madly or not at all or anywhere in between. Now that could have any relationship to whether there's like actually heat or something. But so that's whether the three people in the room are telling the right information. But then there's the attention to that. So if there's precision, if we have precision on the A matrix, which still could be ambiguous or not, then beliefs about hidden state can be updated by observing data more. So it's kind of like scaling the the import the the impact on the basian updating of the beliefs about hidden states from attention to the observation. Um can you elaborate on the row of attention in relation to precision again please? Yeah, I think um attention and precision and temperature sometimes are used in sort of similarish ways. Um, but even like the attention in the transformer, like this is kind this is kind of a uh superb built out version, but in its most um single parameter setting, it's just like how much impact on on the latent state is is being influenced influenced by the new observation coming in. So like complete attention is updating all the way to the new data point. Like someone says it's four and go it's four and then it's five. Okay, it's five. So it's now that could have whatever variance that's whether the object level distribution is just has a given var small or a large variance or whatever or it's a or it's a um dro delta function so it's just single parameter like there's no variance um and then no attention which is kind of one of the one of the multiple places with attention confidence belief where it's like a psychological term moved into stats. Now coming back to confusingly either apply or not apply to that psychological phenomena or be some other psychological phenomena. But it's like if something's not paying attention at all, the other extreme, it doesn't have an impact on it learning. It just it was it was functionally didn't happen. So that's like if you have a um you have some a matrix um the ambiguity matrix mapping observations to latent states. So it could be like a uniform distribution which just is is like the symbols coming over the channel don't reduce uncertainty about the hidden states or it could be like an identity matrix with just the symbols coming over the channel have complete concordance to this latent space um or or other matrices in between. So that's about like the kind of object level properties of the mapping between the observations and the hidden state in the values of the A matrix. Um then that could just have sort of a a implicit kind of attention of one. If if you wanted to explore different impacts of that mapping on belief updating or of modulation within a model of how even um uh those beliefs impact updating. Um having another kind of like a volume knob parameter that can go between this channel is not having any impact to this channel has a high impact is kind of a common mechanism. So sometimes it's done with like a 0ero to one just type scaling, sometimes it's done with like more of a just other ways. And then even though it's a little bit different, here's where there's like a temperature um just scaling different variables because that's kind of one of the ways when it gets down to the calculation of it. That's that's like how these matrices can come to have different impact on each other is just through scaling them like by a coefficient or raising or lowering them to a coefficient. Thank you. Multiplicative weight is a coefficient. Yeah. I guess in terms of math it it can be understood more intuitively. Um so is it is it fair to kind of understand this as I don't know the the update will be reducing uncertainty and in our beliefs about hidden state can be updated by observing data more in that it's kind of like you become more certain about something. Hence you you become more confident in that belief and that's why it's more than if we are no confidence in those beliefs. I think it's the belief in that stream of data is is does that help because I think that's what I was getting kind of tripped up because I just kept thinking okay there's a generative model that is modeling you know whatever that hidden state is and so I'm pretty sure I know what that thing is and then so why do I need all the precision of that channel but it's almost like I am more I think this is really about a a belief or a prediction maybe a model of whatever that is coming from that observable data as opposed to whatever that thing is. And I don't know if that makes sense. So it's so it's like confidence in whatever I'm going to get next. It's almost like the this if there is a model here, it's like it's a model of whatever is passing that message up. And I could be wrong about that, but I don't know if that helps or if it's accurate. Yeah. just one one to Sylvia then than then Andrew. So here with the multiplicative updating on prediction errors. So it's like let's just say that that we're we're we're two two entities are in a zero to 100 action belief space which could be a belief about action like how likely they are to do something. So that's a huge benefit of having direct beliefs about action as opposed to like a reward function is talking about latent states could be talking about a sensemaking latent state could be talking about an action inference. So that's one benefit that's that's not really mentioned here. So now the observation comes in like it's it's a temp it's it's a temperature 0 to 100 and then there's a 0 to 100 action space for a heater. And so one entity might have a um uh the the thermometer reading comes in five units over their belief. So just at the pure error calculation, they're five points off. But then one agent's attention to those five kind of raw units of of like that those are the units that are kind of getting pumped through the A matrix because it's literally where the observation from the thermometer is meeting the latent state beliefs about temperature from the agent. But now at the attention level one of those agents could just turn that five into a.5 by having 0.1 attention. another agent um or another context for the same agent could have a 10 attention. So then it's like 50 units of surprise. So that's the difference between the same five thermometer units being for the purposes of basian updating like it could be like a five a five sigma update or a 50 sigma update in terms of how surprising the event was. Andress. Yes. Uh I'm thinking intuitively like about a scientist with a hypothesis and and this notion of confidence I and I would also relate that to specificity like um confidence can mean two different things. Confidence can mean like a moral state you know like are you taking a stand or are you not? But I think here first of all it doesn't mean anything like that. It just simply means given all the options um how specific are you you know that it is a certain way and not other ways. So like when things are very vague that's in certain sense that's not having confidence in anything. It's just saying you know or it's not specific. But if you're a scientist with a very specific hypothesis that that can be inferred that well your confidence is in that very high and your confidence in everything else is very low. So when you have a very clear specific hypothesis you know quote unquote that you're confident in then um you're really subject to updating because there is something to update. But if you don't have anything specific, you don't really have uh implicitly any notion of confidence uh and there's nothing to update u no matter what you discover because you haven't taken any position or any stand. So that would be I think if if understanding confidence just simply in this kind of default way then the whole other notion would be well what would it mean to have an extra notion of confidence where you can choose to be more invested or less invested but then that would be a whole separate question. I don't think that that's the matter here, but that great example, Andreas, and a great example. So, it's like there's there's there's two rooms. One of them has a uniform distribution of temperature. One of them has a hot spot. Now, there's two investigators. One of them has a uniform belief about the space. The other one has this hotspot concept like they're fitting a Gaussian or they're fitting a DRock delta function to the room. So it's like in the in the in the flat temperature room that sort of flat hypothesis um is the right one and there's no reason to pay attention. But even paying attention or not in that room would keep the keep it the same. Whereas in the uniform room, if there's an investigator who is going to really be paying attention to their updating, then they're going to they could be updating. They could just be wildly moving that small Gaussian around because they they had high precision in in so at that level of updating. So it's just like that there's all these all these kinds of examples help separate out like there's like the volatility of the environment, the volatility of the updating, then there's like the actual dynamics of the environment and then the um the relevance um or the the accurate or veritical nature of the agent's perception. Um or this or this kind of question of like does it learn other features like the whole kind of Hoffman angle with just the evolutionary game theory of of of learning um cognitive structures that you know surprise surprise are not the exact structures in the world. Yeah, those examples make perfect sense now. Thank you. Thank you for the explanations. it becomes very clear with those examples and put that into context. Thank you. Yeah, it it um there's a lot of statistical degrees freedom even in a in a linear regression type even just in the model itself, not even getting into the kind of like hypothesizing after results are known like the HARK um type like iterative inquiry or just modifying aspects of the statistical model, But just within like the POMDP that um from from chapter 4 or within a graph with not that many variables like this, it's like first just the which which pieces are on the board is already a choice. Then there's all the connections and then those connections can be structured different way. And then those connections could be kind of just taken in their plain first order sense. But in a basian setting, if you have just a plain first order variable or matrix, it's like a constant. So that might be fine, but then it it is not necessarily learnable. And even if it were a constant, there still might be another more attentional variable connected to it. One benefit of of basian models in this setting just yeah last point then MB is in um hierarchical parametric linear modeling like explored in uh SPM you can only test uh strongly models that are like strictly nested with each other. So you can test if you do the ANOVA just normal scientific statistics like you have the fMRI data and then you have their age and their their weight and then you can like test for you know factor A factor AXB factor AXB XC um you you can test those as nested ANOVA and then you can say we found evidence for an interaction between the fMRI and age but we didn't find it for weight. So that's that's the way you can do the structural hypothesis testing within parametric classical stats. In basian stats, you can use baze factor and other just sort of empirical um methods to just select models based upon their fit. And they don't need to have a structural relationship to each other. Whereas in the linear statistics, it's like it's more constrained, but it's a more simple test. And that's that's why there's just so much similarity in in ANOVA methods. And the SPM textbook has a lot of this information. Uh MB Thanks. Yeah, I am just curious about the model itself. So to use Andrew's example, which I thought was great, I'm a scientist and um I have a high degree of confidence in my data, right? And so I have a hypothesis and my h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h h hypothesis is confirmed by the data. And so I can see how there's high precision, but there that would be a to me that would seem like a very low model update as far as the the magnitude of the update. Is that am I thinking about that in the right way? Yeah. Okay. So then I could also see high precision ambiguous view. I'm sure that it's ambiguous. Or you could have a high precision low ambiguity view like I'm sure it's this way. Could I also I I could also see how I would have as far as the general model about whatever that hidden state is. I could say I have no idea what that hidden state is, but whatever comes in this channel here, I'm going to believe. So that's that's like a probably a huge belief update, right? Like I have no idea what's coming. Whatever it is, I see I'm I'm definitely buying it, right? I believe it. Yeah. So I guess it doesn't have to do with the you do slot machine. You really don't know if you've won, but the first observation that comes in on on roulette, you're just going to have 100% updating to whatever it is. Uhhuh. So, so I guess it's not really about the certainty of the model itself. It's the certainty of the data. And I think that's the whole the whole point of this, right? We have very precise beliefs about how data are generated then. So it's really not necessarily about the the uncertainty of the models. It's that second level prediction. Is that right? I think you're you're raising the sort of human model interface of how much trust is in that model or um or attention to that model. But yet within the model what the attention so not the not the human's attention to this model's output. Mhm. Just within the model it is it's it's like a scaling coefficient on on blurriness of input where you can blur a channel to just basically be in the backdrop of nothing or sharpen it to to a high layer. Am I jump? Yeah. Yeah. Go ahead. Yeah. MB. Uh I think I disagree with you. Um and the first thing is that I think there's two completely different notions of confidence at play here. One that I think it's natural for us to be injecting, you know, which would be like human personal belief. I don't think that that's implied here. I think all they're saying is that when you have a distribution, because they're talking about any distribution, when you when as soon as you have a distribution, if it's peaked, you're implying that you have confidence in that particular zone as opposed to the other zones. If it's flat, you know, horizontal, then it just implies confidence in any. I mean, you know, just the shape of the distribution indicates a certain type of implicit confidence at play which has nothing to do with mental states or you know like our personal beliefs. It just it's evidenced by the activity you know if it's evidenced by the distribution. So if you are a sense and if you have just intuitively to try to understand like like if you have a clear hypothesis you can be wrong. If you don't have a clear hypothesis you could never be wrong. You don't have a problem with being wrong. If you if you're not going to be wrong you don't need to be updated. If you can be wrong you are at risk of being updated. and you know so so I think that's the that's the issue there is that the more clear and specific you are uh the more you will be open or vulnerable to being updating yourself. The more vague you are, the more kind of like you frame what you say, the more you just kind of make it all mean nothing, the less that you have to worry about updating. Al although being very precise in your beliefs but having no attention to to input does not lead to updating either. So it's not simply that having having precise belief. That's why there's these two notions of confidence at play which I think has to be first you know one is and so the idea of attention see attention works with both versions of confidence but I think the version that they're talking about attention is simply the fact that if you focus on anything you know without having any mental apparatus like a human being but any kind of system that has some kind of focus is going to as soon as you focus you have a implicit notion of confidence uh and If you aren't focused, you have an imp implicit of not really having confidence. So, and the implicit implicit between the discrete world and the continuous world, you know, like you have these continuous actions, they have a notion of confidence, you know, like a specific action, you know, that was taken uh where there's no mental belief, but it's a specific action. But also like in modeling that with our generative models or our like finite models like we have also can have a notion of a specific um thing or maybe maybe even a confidence value. I don't know but I'm trying to deconfuse. I don't know if I am. Yeah, I I I hear you on that. I think there's there's there's probably ways to to write out and program all these things. And again, there might be situations where these models have the same um directional just they capture the variance or or there could be some contrived example or some pattern where they capture different elements. But it's like but again that that makes it clear. There's the investigator with precise or imprecise beliefs about the room, about whatever their hypothesis is actually about. And then they could be paying attention to the thermometer or not. And then that but that can look so many different ways just depending on what the state spaces are because also like beliefs on an information channel. It is a latent state. Like if we have it's like we're we're going around the room with a thermometer and then there's some some unobservable in the room which relates to how variable the thermometer is. That's like the uh recurrent switching linear dynamical system in the RX infer. It's like there's a zero or one. There's two underlying discrete dynamical states. Two, just two states that just are doing this function and then they switch between a low and a high variance mode. But you're only getting these observations just as um raw data points. And then this could this could um just as the variance estimator changes you can infer that you're in a different latent state. So these aren't so different and partially it's because people use latent state to sometimes mean just environmental states but when you get into the model it's it you it gets more specified. Sylvia. Yeah, I just I thought of an example just from a psychiatry computational psychiatry point of view. If we think about people with maladapted beliefs or some sort of psychiatric condition that made them have like negative beliefs about the world and more often than not their beliefs are very very precise. is too precise even though it's inaccurate. And the reason that they are not really updating that belief is that they they will actually purposefully ignore certain data that would challenge that negative model, the negative belief or they will selectively just choose the things that will confirm that model and that's why they they will appear to be trapped in in that belief. I guess I guess that would be an example um for the differentiation that you mentioned about on like whether you pay attention or not um to to that data. Yeah, there there could be a lot of ways that in a in a complex model it plays out, but like precision on how how informative 0 to 100 beliefs on the screen are taken to be and then there's like the the epistemic foraging of choosing where to look. Whereas in a lot of these textbook type examples, it's like the the agent is just kind of locked into the one stream of information and we're just talking about the dynamics of one stream of information coming in like the the sequence of music notes. But once you start getting into like behavior switching or location switching, that's that is whole different spaces of what information to choose. But it's like everyone selectively chooses information and and some would probably even extend that some that that all selectively choose information to confirm their model in the kind of self-evidencing maximalist take. um John and then address. Um if you if you talk about the the uh information coming in um you have to somehow figure out the mutual information between what what you're seeing and what you want to know and that in general is a very hard problem. Uh so how would you how do you evaluate that someone else has um you know uh uh what's it called? Um psych uh I don't want to say psychopathology because that means something specific but uh pathologies of the mind. Uh how how would how can you tell from the outside because you you have to you have to know the in mutual information and that's typically very very hard. Yeah, short answer um empirically the mutual information is going you're going to call some computer programming like package like mutual information between these two distributions. So there there's more to um like what is the actual mutual information or what is actually happening but just at the sort of really pragmatic level mutual information is also going to be conditioned upon what model you have like if your if your model is just a coin flip then you're only going to be able to get one bit of information if you are maximally imprecise about your belief whereas like you know if you had different beliefs about the coin flip then you might get a amount of information from it. But that shows just quantitatively um that the it the mutual information is about the distributions in the model. So then you can look and infer different parameters in your model. You fit your model to data and then you look at the parameters of your model and their variance profiles. Whether you do it with classical like parametric statistics or basian statistics or um non-parametric like sampling based methods you're going to get some graph that looks like this. This could be like the risk score of five different properties. This could be that some cognitive parameter for five different entities or five different contexts. So that could be wrong, but that is your um calibrated and empirically driven belief about a latent parameter reflecting some cognitive phenotype of interest that you explicitly specified, right? But I I just don't see how you get out of this situation of uh your model's pathological. No, your model's pathological. You don't have the mutual information. No, you don't have the mutual information. Oh, yeah. Yeah. I mean, there's two Yeah. Okay. So now we have two two um modelers meet and they're they're drafting the DSM or they're drafting whatever and then they they um and then even on the same model one could say well the threshold for this or that should be 10% or it should be two sigma it should be this. It's like debating where's the threshold and what to do with model outputs for the same model. That's a thing and then debating which model that's there's that is another space of discourse but just within a given models's output those yields beliefs on latent state but what what you do with these risk estimates or what how you made and how you selected your profile of risk modeling models That's That's not just like you just make one and just take it. Um Axel then Andreas. I think Andreas was first so you can Okay. Yeah, go ahead.

I'll jump in. Um I think um Sylvia's example relates to the whole basil ganglial issue about uh how can action be changed and I think that belief changing may be very similar like to changing the state of action and this would relate to John's uh issues regarding you know objectively talk about pathological thinking. So active inference is very incomplete in many ways. But with regard to this specific type of issue, it's suggesting that if you want to change action, you need to lower your precision, you know, in terms of your understanding of what the current state is. Otherwise, you'll never be able to get out of it. You'll be stuck in a rut. And similarly like if you want to change your beliefs, you have to be able to let go of your position and reduce it down so that then you could re-evaluate and reclaim it or to change it. And so if you don't do these things, you'll never be able to change your state of action. You'll never be able to change your uh beliefs. And it could be said like there's a pathology in that in the framework of active inference. And there's probably I mean there's examples I think of where you can see that neurologically but uh it's it's just a general mechanism. Yeah. But but there there are approaches where people say the fact that I'd ever update my beliefs is what makes me right. And you know uh trapped in the world of probability. It's hard to well there's a virtue to having very strong beliefs. So let's say I believe in God, but if I insist that the name of God is, you know, uh, Joseph on Thursdays and Mary on Fridays, you know, like, well, am I willing to let go of that belief and reclaim it? You see, like not let go, but like to attenuate it, you know, if if a person's not able to attenuate it, that means that they're not able to change beliefs. Then it's not clear why to have beliefs, you know, like why be alive in the first place. So I think it's so it's just saying like if you accept active inference as a theoretical framework why um well but I think the idea is they would have the beliefs because they are there are preferences that are adaptive right there just a lot of pieces again it doesn't necessarily need to apply to this like self-standing of life a at all this just can be understood and and possibly more formally accurately just at the level of variables in the relationships, but there are some intriguing parallels. But just I I don't think every single analytical relationship needs to um map to any given psychological one when they're simple motifs, but those are all examples of things that deploy this simple motif in in intense ways. Axel. Yeah, thanks. Uh so I have a question which is much less related to chapter 5 and also less related to these uh philosophical discussions. So I don't know if it's actually better to bring up in another uh meeting if it's more like practical implementation of active inference. Uh but yeah I can ask if no one else has Yeah, just go for it. Sure. Maybe if I could share my screen would that be possible? I can just show what the problem is. Sure. Go for it. There there is only but yes go for it. Yeah sure. I go fast. So basically uh I'm writing my master's thesis right now and I'm very interested in trying to combine causal discovery with active inference. So causal discovery is basically you know you have uh different nodes. You could have a causal graph that says X causes Y which causes C and you're trying to find out the true causal structure. uh there's a lot of literature on this but I thought it was specifically interesting to use active inference for this maybe specifically for intervention selection. So for example let's say you have all kind of different variables and you have to figure out how they um relate to each other what causes what. Sometimes you'll have to intervene on certain variables in order to learn something. So kind of with this example here, you uh wouldn't actually from observational data be able to distinguish between, you know, let's say that X and Y and C are all co-bearing, but you don't necessarily know if it's uh Y that causes X and causes C or this other one. So in order to figure that out, you have to carry out some interventions. And I'm kind of trying to do that with active inference right now. And I was hoping that maybe uh some of this kind of uh reducing uncertainty um and minimizing free energy could be used to maybe choose the interventions that were the most uh um like reduce the uncertainty the most and that would be interesting. I've set up kind of a simple example here where I have um a true causal graph which is this one. And uh here my states have become uh all the possible uh values that these different variables can take on. And I have some um probabilities over actions. And I I start out by just having my B matrix uh complete. My A matrix basically uh is known. It it says okay when the observations are the same as states when all these variables are 00 0 I will see that they are 0 0 0 and then I have my probabilities over actions where I am able to do no operation so I'm not intervening at all I can intervene on x this is basically like toggling it so if x was zero and then I intervene on x it becomes one and so on and I run through this example and I have my uh my um c like I I start out my prior D. I start out in this 0 0 state. My desired state is to be in this uh second to last state where X is one and Y is one and C is zero. And then I kind of start out and I run it. Uh and it says okay um you know it starts out by having a equal probability of all actions. I try X. This takes me to the final state because X causes Y to be one and causes C's to be one. Um and let's see here. Now I'm in this final state and again I have equal probability over reactions because I've never been in this state before. And maybe just to go a bit quicker over it now because we don't have that much time. What ends up happening is that I keep getting into these different states when I haven't been in a state before. I haven't learned anything about where my actions will take me in the B matrix. So therefore I'll have even probability. It takes pretty long time before I finally get around to getting to the uh let's see here the desired state which is have it here. So I get to the desired state here and now you'll see that because it's okay. Now this is of course always with demos it doesn't work but basically I end up getting to the desired state that I want which is um which is this state right before and then I just keep doing no op. It takes me a very very long time to get to this place and I'm wondering if just quickly and maybe anyone else also has some ideas on this whether active inference has been used for this kind of causal discovery before and whether one could learn the causal structure in less time because right now it seems that my big uh problem is that I I maybe learned that no op doesn't change anything but I only learn that in the first state and then I have to also learn that it doesn't change anything in the second state and the third state and the fourth state and so on And yeah, I was just interested in any input on that and any thoughts if anyone had some. Yeah, great work. Definitely cool to see that kind of direct active for the causal discovery. um when you are initiating these new um slices. Mhm. There are horistics that could um accelerate and transfer the learning like uh I if uh if we have a room with all these locations and then we have the heater in the cooler, we don't need to learn that heater we don't need to have a flat prior on what the heater does at every single square. So like instead of initiating as a flat um belief when you were bringing in these new blue ones, they could just take the average of the other ones. Okay. But there's kind of no free lunch because you could then you would be you could there's a situation where that will learn fast if the relationships are transferable. But if they are then doing not initi initializing the matrix with like the rows or the column or whatever averages or just or or however other just initialization heristics can go really far. Would I be able to find this somewhere? Has this already been this kind of averaging or you just coming up with this now? Um I don't I don't know what specific paper but I've seen it in just regular statistics and ways of doing okay interesting and and then just maybe one final question. So another way right now the way that I'm setting it up is that I'm uh basically the states that I have are the true combinations of all values that there could be. Another way of framing this is that instead of doing uh states as the values, I actually do the hidden states as the true causal structures. Um so for example, I say one state is X causes Y, the other is that X is caused by Y. And maybe they're independent of each other. are the three different possibilities in this example. But if you were to do this, then my hope would be that you would finally learn the true hidden state, right? Which would be the true causal structure. H however, I'm a little bit unsure about what would the observations be for this, right? Um especially because the observations in the data would be the same for or at least this and the first one and the second one. That's like the simpler MDP case. Like if you want to just do away with the partially observable part and the whole observation to latent state mapping, it's like just working within the latent space because observations and the and the latent states are the same. Sometimes people do that. Other times um do still have the A matrix but just make it an identity matrix. So then you still get to use observations and have a separator which could later be fuzzed but you can start with just passing it through basically right and would the B matrix then be useful at all because I'm thinking for example here I mean the way that I would learn like let's say um the way that I would learn that X causes Y instead of the Y causes X is that I would uh well this is a very sick I would carry out an intervention on the X. So I say like do x set it to one and then maybe I would see that if y changes when I change x when I carry out this intervention then I would guess that I'm in this state. If y doesn't change when I change x then I can infer that either I'm either they're independent of each other or y causes x. Um but then I feel like I would have to create an a matrix for every single intervention. Then I have to say okay if I have uh if I try out an intervention on X and I see that uh X and Y are independent uh then I know that I'm in one of these two final states. If I see the Y changes, I know I'm here. But that it seems I'm getting a little bit too far away from what active inference is. If I'm creating like many many different A matrices or what do you think? Well, one one answer is it part it just depends on whether you're continuing to do a variational um or or expected or other free energy. Like if you're still doing policy inference with EF on the B then these are just and then it's just what question are you really trying to ask like are you in a situation where the causal structure is fixed and the agent is inferring it or can there be switching of the causal structure I mean or or just what or can the agent act to change the causal relationships or only just kind of poke each of them to give an output. And that's kind of like the dynamic causal modeling setting that again SPM kind of led to which is taking in noisy EEG fMRI data and then asking what is the uh effective connectivity in a structural dynamical causal model of these brain regions like is there any causal influence of the of the uh implicit neural activity proxied by the bold on these two regions. If so, what is the dynamical structural model for them? And then they show in the textbook and like this is kind of like why I come back to it because it shows here's how it's done with generalized linear modeling with parametric classical stats. Approach two, here's how it's done with basian statistics. Here's here's pros and cons of doing variational inference. Here's here's pathologies and heristics for priors. And then third way, here's how it's done with non-parametric statistics like bootstrap, reample, which are like non-parametric there. But so just the the all those tools really well worked for the brain imaging setting where there's this like highdimensional noisy lagged proxied fMRI measurement. Mhm. But does it do you feel like it intrinsically is there an intrinsic motivation in the active inference framework or drive to learn the causal structure? Because in the examples I showed you before, I was telling it that I uh that my preferred observation was to be in a certain state or have a certain observation. In reality, I don't have a preferred observation. I have my preference is that I find the true causal structure. Right? So it'd be very hard to kind of say that there's a certain observation I want to see and that's why I'm trying to maybe change the framing a bit but I don't know maybe John he has some yeah last comment but yeah I hope you more and we can discuss it more question for Axel actually. So these X and Y are these propositions or are these like states in a phase space? Right now they're just states right now they're binary. So x can be 01, y can be 01. Very simple. So the so so they're boolean. So they're propositions, would you say? Yeah, you could say that. Well, it looks like y x implies y, right? But it is kind of saying that x is causing y. So if x is one, y will also become one. Yeah. And and there's there's just there I'm gonna just end it just because it's we're already past the time, but there's a lot of a lot of cool work with causal nets and petri nets and all these kinds of things. So, thank you though for sharing that. All right. Yeah. Thanks so much for the input. Thank you all. Bye.